

Project 02

AI Music & Audio Analyzer

Test & Development Report — v0.9

Item	Current Value
Project Status	In Progress — Steps 1–17 validated
Platform	Windows Laptop
Python	3.14.7
Environment	VS Code + PowerShell + Python venv
Hardware	Windows Laptop + Airdopes Joy Bluetooth Earbuds
Music Input	test_music.mp3
Current Milestone	Step 17 — Audio Quality & Health Score
Overall Health Score	96 / 100 — Excellent

1. Project Objective

Build a practical Python-based AI Music & Audio Analyzer using a Windows laptop and Airdopes Joy Bluetooth earbuds. The project progressively validates Bluetooth audio capture, digital signal processing, spectral analysis, frequency-band analysis, rhythm analysis, musical key estimation, loudness analysis, clipping detection, visualization and explainable audio-health scoring.

2. Project Architecture

Airdopes / Music File → Audio Loading → Signal Level → Waveform → FFT → Spectrogram → Spectral Features → Bass/Mid/Treble → Tempo/Beat → Integrated Summary → Key/Chroma → Loudness/Dynamic Range → Clipping Detection → Audio Quality Score → Visualization

3. Completed Steps

Step	Feature	Status
1	Audio Device Detection	PASS
2	Airdopes Microphone Verification	PASS
3	Airdopes WAV Recording	PASS
4	WAV Audio Analysis	PASS
5	Waveform Visualization	PASS
6	Audio Level Analysis	PASS
7	FFT / Frequency Spectrum	PASS
8	STFT / Spectrogram	PASS
9	Spectral Feature Analysis	PASS
10	Bass / Mid / Treble Energy	PASS
11	Tempo / Beat Analysis	PASS
12	Integrated Audio Feature Summary	PASS
13	Real Music File Input & Analysis	PASS
14	Professional Music Visualization Dashboard	PASS
15	Key / Chroma Analysis	PASS
16	Loudness / Dynamic Range / Clipping	PASS
17	Audio Quality & Health Score	PASS

4. Hardware Audio Validation

Parameter	Result
Device	Airdopes Joy Hands-Free
Input	1 channel / 44,100 Hz
Recording Duration	5.00 sec
File	audio\airdopes_test.wav
RMS Level	-30.15 dBFS
Peak Level	-7.45 dBFS
Crest Factor	13.638
Signal Classification	Healthy

5. Real Music Track Baseline

Parameter	Result
File	test_music.mp3
Duration	240.02 sec
Sample Rate	44,100 Hz
Channels	1 (mono)
Samples	10,585,088
RMS Level	-17.05 dBFS
Peak Level	-0.17 dBFS
Bass / Mid / Treble	89.42% / 10.50% / 0.08%
Dominant Band	Bass
Estimated BPM	109.96
Detected Beats / Onsets	437 / 1,052

6. Key / Chroma Analysis — Step 15

Feature	Result
Dominant Pitch Class	D
Estimated Key	G Minor
Best Correlation	0.7472
Second Candidate	D Minor
Second Correlation	0.7196
Confidence Gap	0.0276

Feature	Result
Reported Confidence	High
Output	reports\music_key_chroma.png

G Minor is the strongest candidate from the chroma/profile correlation method. D Minor is a close alternative, so the project reports G Minor as an estimated key rather than ground truth.

7. Loudness / Dynamic Range / Clipping — Step 16

Parameter	Result
RMS Amplitude	0.140450
RMS Level	-17.05 dBFS
Peak Amplitude	0.980979
Peak Level	-0.17 dBFS
Crest Factor	6.985
Peak-to-RMS Dynamic Range	16.88 dB
Average Short-Term RMS	-18.07 dBFS
Maximum Short-Term RMS	-10.10 dBFS
Minimum Short-Term RMS	-102.59 dBFS
Short-Term Level Variation	92.49 dB
Clipping Threshold	0.999
Total Clipped Samples	0
Clipping Percentage	0.000000%
Clipping Status	No Clipping Detected
Output	reports\music_loudness_dynamic_range.png

The 92.49 dB short-term variation is strongly influenced by very low-level/silent frames and is not treated as a standardized music dynamic-range measurement. The 16.88 dB peak-to-RMS value is retained as a simple signal-level dynamic indicator.

8. Audio Quality & Health Score — Step 17

Step 17 combines existing analysis results into a transparent, project-specific 100-point Audio Quality & Health Score. It is an explainable engineering heuristic, not an industry-standard mastering or perceptual-quality metric.

Component	Score	Maximum
Signal Level	20	20
Clipping Safety	20	20
Dynamic Characteristics	15	15
Frequency Balance	11	15
Spectral Quality	10	10
Rhythm	10	10
Key / Chroma	10	10
TOTAL	96	100

Final Classification: **96 / 100 — Excellent**

Project Finding	Result
Estimated Key	G Minor
Key Correlation	0.7472
Estimated BPM	109.96
Dominant Frequency Band	Bass — 89.42%
Clipping	0 samples
Peak Level	-0.17 dBFS
Peak-to-RMS Dynamic Range	16.88 dB
Health Classification	Excellent
Output	reports\audio_quality_health_score.png

9. Professional Visualization Dashboard — Step 14

The initial dashboard was refined into a clearer professional v2 layout with KPI cards, full-track waveform, logarithmic frequency spectrum, frequency-balance bars, focused spectrogram and a dedicated spectral/rhythm information panel.

Output	Purpose
music_analysis_dashboard.png	Original dashboard retained for project history
music_analysis_dashboard_v2.png	Refined professional dashboard
music_key_chroma.png	Step 15 chroma and key visualization
music_loudness_dynamic_range.png	Step 16 loudness and clipping visualization
audio_quality_health_score.png	Step 17 quality and health score visualization

10. Current Project Structure

Project-02-AI-Music-Audio-Analyzer/

- audio/ → airdopes_test.wav
- music/ → test_music.mp3
- reports/ → waveform, level, FFT, spectrogram, spectral, band, tempo, dashboard, key/chroma, loudness and health-score outputs
- src/ → Steps 1–17 Python analysis scripts
- tests/
- venv/
- requirements.txt
- README.md

11. Overall Validation

Validation	Status
Airdopes Bluetooth microphone accessible	PASS
Airdopes recording	PASS
WAV signal analysis	PASS
Waveform / FFT / spectrogram	PASS
Spectral features	PASS
Bass/Mid/Treble	PASS
Tempo/beat analysis	PASS
Integrated analyzer	PASS
Real music input and analysis	PASS
Professional dashboard	PASS
Key/chroma estimation	PASS
Loudness/dynamic-range analysis	PASS
Clipping detection	PASS

Validation	Status
Audio Quality & Health Score	PASS
AI interpretation	PENDING

12. Step 17 Interpretation

The track achieved 96/100 and was classified as Excellent under the project's scoring rules. Full scores were achieved for signal level, clipping safety, dynamic characteristics, spectral quality, rhythm and key/chroma confidence. Frequency balance received 11/15 because the track is strongly bass-dominant. This does not mean the music is objectively poor; it means the project's scoring model recognizes that fewer frequency regions contain significant relative energy.

13. Engineering Limitation

The Audio Quality & Health Score is an explainable heuristic created specifically for this learning project. It should not be presented as a professional mastering grade, LUFS compliance result, codec-quality score or standardized perceptual-quality metric. Future versions can add calibrated loudness measurements and perceptual models.

14. Next Development Step

Step 18 — AI-Based Music Understanding. The next phase can move from signal measurements toward higher-level interpretation, such as mood estimation, genre characteristics, instrumentation indicators, speech/music classification and natural-language analysis of the track.

15. Version History

v0.1 — Project-02 documentation baseline created from the beginning.

v0.2 — Steps 4–6: WAV analysis, waveform visualization and audio-level analysis.

v0.3 — Steps 7–8: FFT spectrum and STFT spectrogram.

v0.4 — Step 9: spectral feature analysis.

v0.5 — Steps 10–11: Bass/Mid/Treble and Tempo/Beat analysis.

v0.6 — Steps 12–13: integrated feature summary and real music track analysis.

v0.7 — Step 14: professional music visualization dashboard and refined v2 layout.

v0.8 — Steps 15–16: key/chroma analysis plus loudness, dynamic range and clipping analysis.

v0.9 — Step 17: transparent Audio Quality & Health Score; final result 96/100 — Excellent.

Only Project-02 development history is included.